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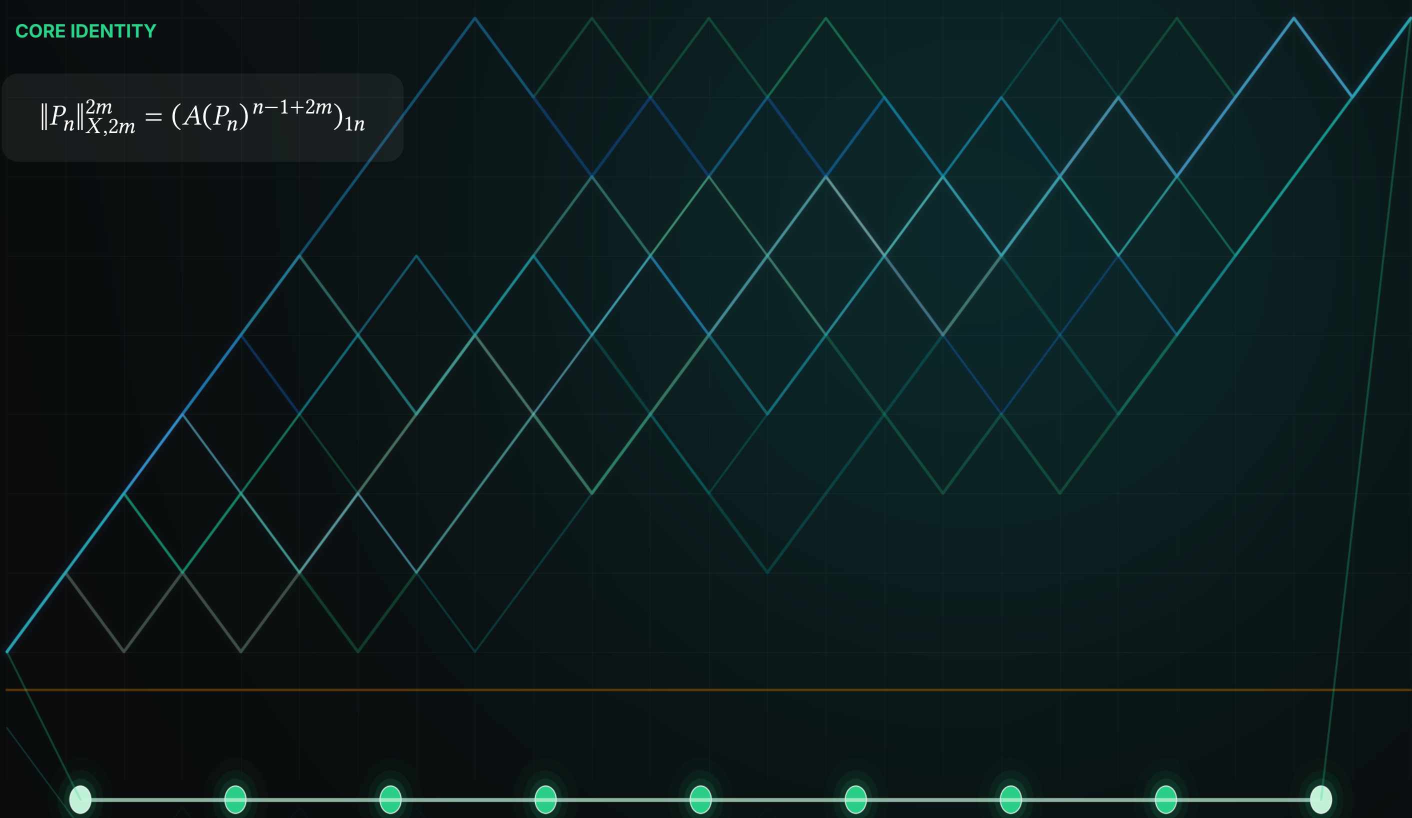
SPECTRAL GRAPH THEORY / ALGEBRAIC COMBINATORICS / RANDOM VECTOR NORMS

ENDPOINT WALKS EVALUATE PATH-GRAPH CHS NORMS

A proof of the exact formulas conjectured for path graphs, with corrected finite-support and boundary conditions

CORE IDENTITY

$$\|P_n\|_{X,2m}^{2m} = (A(P_n)^{n-1+2m})_{1n}$$



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Endpoint Walks Evaluate the Complete Homogeneous Symmetric Norms of Path Graphs

A proof of the formulas conjectured in Remark 4.3 of Bouthat, Chávez, and Sheng, with boundary corrections

Metriq PRISM Laboratory
Program for Research in Intelligent Systems and Methods
A research laboratory of Metriq Foundation, Inc. • Florida, United States • metriq.org/prism
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Research status

Bouthat, Chávez, and Sheng explicitly leave the evaluation of the complete homogeneous symmetric norm of the path graph P_n as an open problem and conjecture two equivalent formulas. This manuscript proves the principal formulas through an endpoint-resolvent identity and identifies off-by-one conditions in the finite truncation and short-range simplification. The argument is exact but has not yet undergone independent specialist review or peer review.

Abstract

Let P_n be the path graph on vertices $1, \dots, n$, let A_n be its adjacency matrix, and let $\|P_n\|_{X,d}$ denote the random-vector norm induced by a standard exponential random variable. For even d, its dth power is the complete homogeneous symmetric polynomial h_d evaluated at the eigenvalues of A_n. A June 2026 preprint of Bouthat, Chávez, and Sheng leaves the exact evaluation of this norm open and conjectures trigonometric and binomial expressions. We prove the stronger structural identity

$$\|P_n\|_{X,d}^d = (A_n^{n-1+d})_{1n},$$

so the CHS power is the number of length-$n-1+d$ walks from one endpoint of P_n to the other. The identity follows by comparing the CHS generating function $\det(I - tA_n)^{-1}$ with the endpoint entry of the resolvent $(I - tA_n)^{-1}$. Spectral decomposition gives the conjectured trigonometric sum, while a two-barrier reflection argument gives the conjectured binomial sum. We also derive the exact finite support, show that the published non-strict short-range condition fails at equality, and supply the corrected equality formula. Finally, we give a rational generating function and a linear recurrence for all even degrees. Exact integer verification over 1,176 parameter pairs is included as a reproducibility audit. No claim of independent peer review or publication priority is made.

Main conclusion. For every $n \geq 1$ and even $d \geq 2$, the exact CHS power of P_n is an endpoint-walk count. Consequently, both infinite formulas conjectured in Remark 4.3 are valid. The stated finite-support bounds and the condition for the one-term closed form require boundary corrections.

MSC 2020. Primary 05C50; Secondary 05A15, 15A18, 60C05. **Keywords.** Path graph, graph spectrum, complete homogeneous symmetric polynomial, random-vector norm, lattice walk, reflection principle.

1 Problem context

Let G be a simple graph of order n with adjacency eigenvalues $\lambda_1, \dots, \lambda_n$. If X is a standard exponential random variable and $d \geq 2$ is even, the associated random-vector norm satisfies [1, 2]

$$\|G\|_{X,d}^d = h_d(\lambda_1, \dots, \lambda_n), \tag{1}$$

where h_d is the complete homogeneous symmetric polynomial of degree d. Its generating function is

$$\sum_{r \geq 0} h_r(\lambda_1, \dots, \lambda_n) t^r = \prod_{k=1}^{n} \frac{1}{1 - \lambda_k t} = \frac{1}{\det(I - tA(G))}. \tag{2}$$

In Section 4.1 of their June 2026 preprint, Bouthat, Chávez, and Sheng state that the CHS norms of complete graphs, complete bipartite graphs, and matching graphs have simple expressions, but that they leave the evaluation of $\|P_n\|_{X,d}$ open [1]. Their Remark 4.3 records experimentally supported trigonometric and binomial formulas. The proof below confirms those principal identities and explains them combinatorially.

The central observation is that the reciprocal determinant in equation (2) is already present in an endpoint entry of the path resolvent. That identification turns a symmetric-polynomial coefficient into a constrained lattice-walk count.

Notation used throughout

Let

$$d = 2m, \qquad N = n + 1, \qquad L = n - 1 + 2m = N - 2 + 2m,$$

with $n \geq 1$ and $m \geq 1$. We use the convention

$$\binom{L}{q} = 0 \quad \text{when } q \notin \{0, 1, \dots, L\}.$$

The vertices of P_n are $1, \dots, n$, in their natural order, and $A_n = A(P_n)$.

2 Main theorem

Theorem 2.1 (Exact evaluation of the path-graph CHS norm). *For every $n \geq 1$ and $m \geq 1$, let $N = n + 1$ and $L = n - 1 + 2m$. Then*

$$\begin{aligned} \|P_n\|_{X,2m}^{2m} &= (A_n^L)_{1n}, && (3)\\ &= \frac{2^{n+2m}}{n+1} \sum_{k=1}^{n} (-1)^{k+1} \cos^L\left(\frac{\pi k}{n+1}\right) \sin^2\left(\frac{\pi k}{n+1}\right), && (4)\\ &= \sum_{j \in \mathbb{Z}} \left[\binom{L}{m + jN} - \binom{L}{m - 1 + jN}\right]. && (5) \end{aligned}$$

The sum in equation (5) has exact finite support

$$-\left\lfloor \frac{m}{N} \right\rfloor \leq j \leq 1 + \left\lfloor \frac{m-1}{N} \right\rfloor. \tag{6}$$

If $1 \leq m < N$, then

$$\|P_n\|_{X,2m}^{2m} = \frac{n^2 + n - 2m}{(n + 2m)(n + 2m + 1)} \binom{n + 2m + 1}{m}. \tag{7}$$

At the boundary $m = N = n + 1$, the corrected value is

$$\|P_n\|_{X,2(n+1)}^{2(n+1)} = 1 + \frac{n^2 - n - 2}{(3n+2)(3n+3)}\binom{3n+3}{n+1}. \tag{8}$$

In particular, the norm itself is the positive $2m$th root of any of the equal quantities above.

Remark 2.2 (What is confirmed and what is corrected)**.** The infinite trigonometric and binomial expressions proposed in Remark 4.3 are confirmed by theorem 2.1. Two auxiliary statements in that remark require adjustment after writing $d = 2m$ and $N = n + 1$:

1. the exact lower and upper support indices are those in equation (6);
2. the closed form equation (7) requires $m < N$, not $m \le N$.

When $m = N$, an additional image term at $j = -1$ contributes exactly 1, yielding equation (8).

3 The resolvent identity

The proof begins with a cofactor calculation for the tridiagonal path matrix.

Lemma 3.1 (Endpoint entry of the path resolvent)**.** *For every $n \ge 1$,*

$$\big((I - tA_n)^{-1}\big)_{1n} = \frac{t^{n-1}}{\det(I - tA_n)}. \tag{9}$$

Proof. For $n = 1$ the identity is immediate. Suppose $n \ge 2$. By the adjugate formula, the numerator of the $(1, n)$ entry is the $(n, 1)$ cofactor of $I - tA_n$. Deleting row n and column 1 leaves a triangular matrix whose only contributing permutation selects the $n - 1$ entries $-t$ adjacent to the deleted diagonal. Its minor determinant is $(-t)^{n-1}$, and the cofactor sign is $(-1)^{n+1}$. Their product is t^{n-1}, which gives equation (9). □

Theorem 3.2 (CHS coefficients are endpoint walks)**.** *For every integer $r \ge 0$,*

$$h_r\big(\lambda(P_n)\big) = (A_n^{n-1+r})_{1n}. \tag{10}$$

Consequently, equation (3) holds for $r = 2m$.

Proof. As a formal power series,

$$(I - tA_n)^{-1} = \sum_{\ell \ge 0} A_n^\ell t^\ell.$$

Taking the $(1, n)$ entry and applying lemma 3.1 gives

$$\sum_{\ell \ge 0} (A_n^\ell)_{1n} t^\ell = \frac{t^{n-1}}{\det(I - tA_n)}.$$

By equation (2), the right-hand side is

$$t^{n-1} \sum_{r \ge 0} h_r\big(\lambda(P_n)\big) t^r.$$

Comparison of the coefficient of t^{n-1+r} proves equation (10). Equation (1) then gives equation (3) for even $r = 2m$. □

Since an entry $(A_n^L)_{uv}$ counts length-L walks from u to v, theorem 3.2 is also a direct combinatorial interpretation of the CHS power. Parity is automatic: an endpoint-to-endpoint walk has length congruent to $n - 1$ modulo 2, so odd r gives zero, as expected from the symmetric spectrum of a bipartite graph.

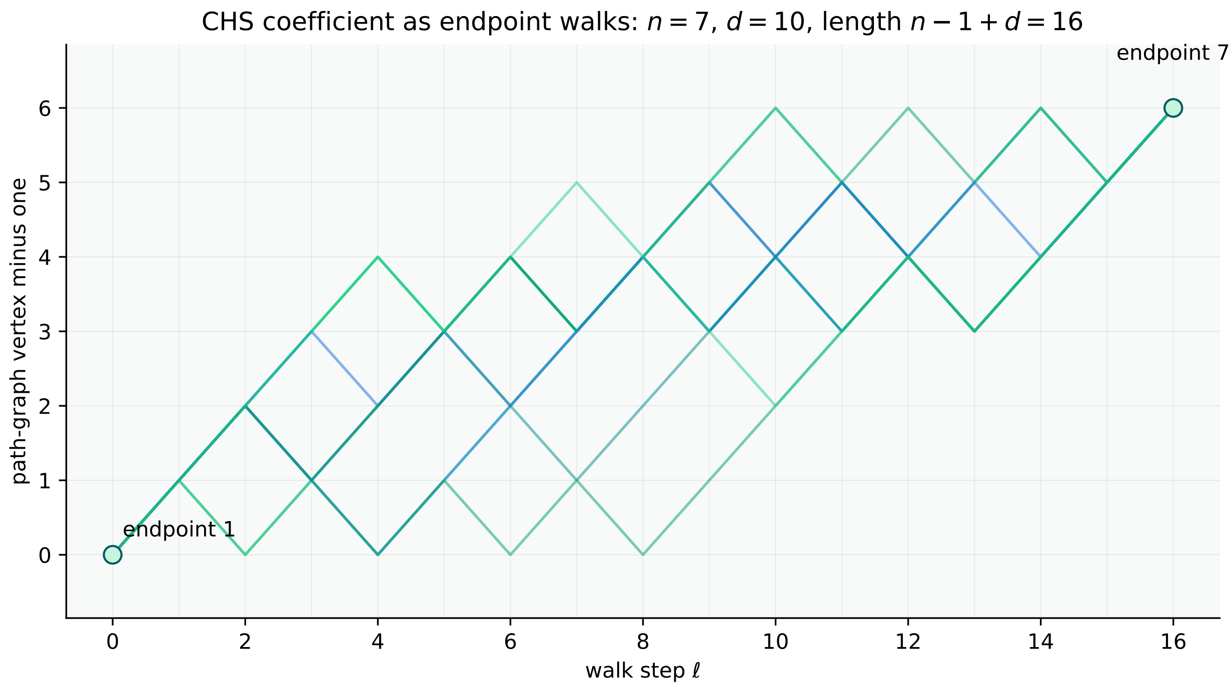

Figure 1: Sample length-16 endpoint walks in P_7, shown after shifting the vertex coordinate down by one. The exact CHS power $\|P_7\|^{10}_{X,10}$ counts all such walks, not only the displayed sample.

4 Spectral evaluation

The path spectrum and normalized eigenvectors are standard [3]:

$$\lambda_k = 2\cos\theta_k, \qquad v_k(j) = \sqrt{\frac{2}{N}}\sin(j\theta_k), \qquad \theta_k = \frac{\pi k}{N}, \tag{11}$$

for $k = 1, \ldots, n$ and $j = 1, \ldots, n$.

Proof of equation (4). By spectral decomposition and equation (11),

$$\begin{aligned}(A_n^L)_{1n} &= \sum_{k=1}^{n} \lambda_k^L v_k(1) v_k(n) \\ &= \frac{2^{L+1}}{N} \sum_{k=1}^{n} \cos^L(\theta_k)\sin(\theta_k)\sin(n\theta_k).\end{aligned}$$

Since $n = N - 1$,

$$\sin(n\theta_k) = \sin(k\pi - \theta_k) = (-1)^{k+1}\sin(\theta_k).$$

Finally, $L + 1 = n + 2m$, which yields equation (4). □

This derivation does more than verify a trigonometric identity: it explains why the alternating sign and squared sine weight occur. They are the product of the two endpoint coordinates of the kth normalized path eigenvector.

5 Two-barrier reflection formula

We now count the same endpoint walks without diagonalizing A_n. For an integer r and $\ell \geq 0$, define the unrestricted one-dimensional walk kernel

$$p_\ell(r) = \binom{\ell}{(\ell - r)/2}, \tag{12}$$

with the convention that the value is zero unless the lower argument is an integer between 0 and ℓ. This counts length-ℓ walks with steps ± 1 and net displacement r.

Lemma 5.1 (Two-barrier image formula). *Let $N \geq 2$ and $1 \leq a, b \leq N-1$. The number of length-ℓ walks from a to b that remain in $\{1, \dots, N-1\}$ is*

$$Q_\ell(a,b) = \sum_{j \in \mathbb{Z}} \left[p_\ell(b - a + 2jN) - p_\ell(b + a + 2jN) \right]. \tag{13}$$

Proof. Only finitely many summands are nonzero. The unrestricted kernel satisfies

$$p_{\ell+1}(r) = p_\ell(r-1) + p_\ell(r+1).$$

Therefore Q_ℓ satisfies the same nearest-neighbor recurrence in the target coordinate:

$$Q_{\ell+1}(a,b) = Q_\ell(a, b-1) + Q_\ell(a, b+1).$$

The image pairs cancel at both absorbing barriers. At $b = 0$, use the evenness of p_ℓ and reindex $j \mapsto -j$. At $b = N$, use evenness and reindex $j \mapsto -j - 1$. Thus

$$Q_\ell(a, 0) = Q_\ell(a, N) = 0.$$

At time zero, the first image family contributes 1 exactly when $a = b$ and the second contributes nothing, so $Q_0(a,b) = \delta_{ab}$. The initial condition, boundary conditions, and recurrence uniquely determine the constrained walk counts, proving equation (13). □

Proof of equation (5). Apply lemma 5.1 with $a = 1$, $b = N - 1 = n$, and $\ell = L = N - 2 + 2m$. Then

$$p_L(N - 2 + 2jN) = \binom{L}{m - jN},$$
$$p_L(N + 2jN) = \binom{L}{m - 1 - jN}.$$

Reindexing $j \mapsto -j$ and using equation (3) gives equation (5). □

6 Finite support and the boundary correction

The infinite notation in equation (5) is convenient, but the exact support can be read directly from the two lower binomial arguments.

Proposition 6.1 (Exact support). *For $m \geq 1$, a summand in equation (5) can be nonzero only if*

$$-\left\lfloor \frac{m}{N} \right\rfloor \leq j \leq 1 + \left\lfloor \frac{m-1}{N} \right\rfloor,$$

and both endpoints of this interval contribute for some parameter values. Hence these are the exact uniform support bounds.

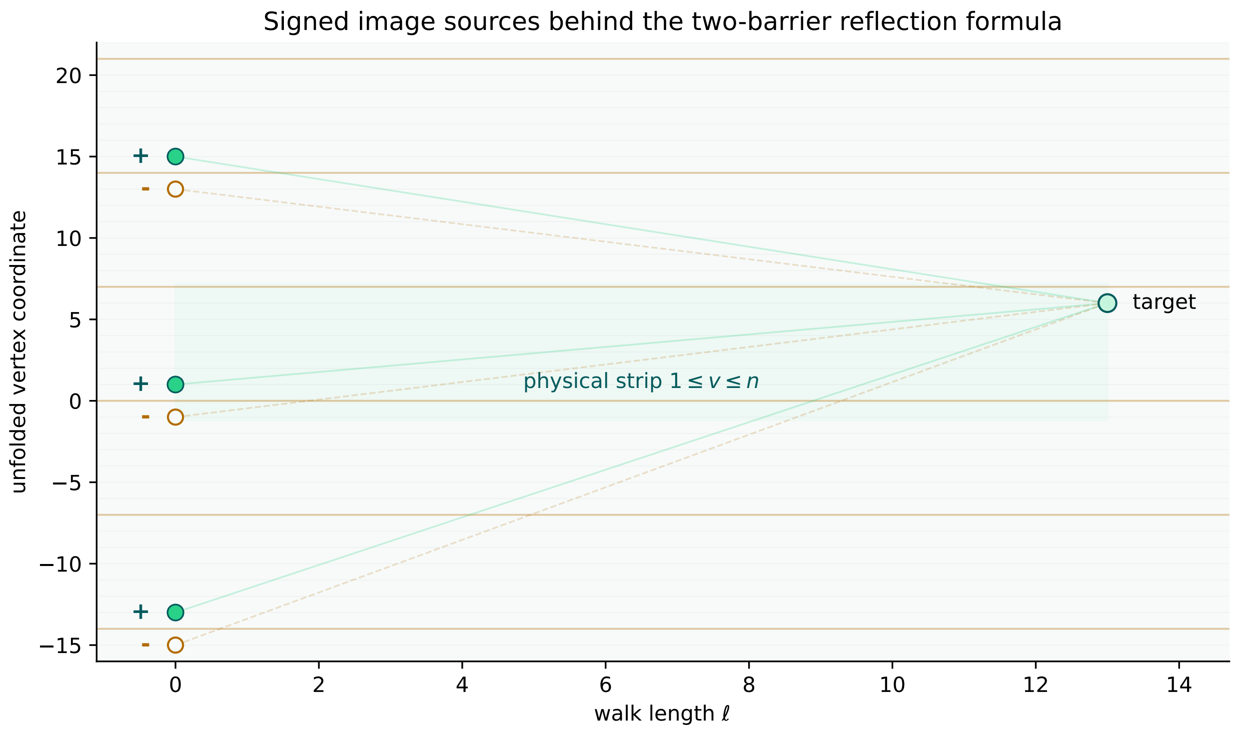

Figure 2: Schematic signed image sources for a strip with two absorbing barriers. Positive and negative unrestricted kernels cancel at the boundaries, leaving exactly the walks that remain inside the physical path.

Proof. For the first binomial coefficient, nonvanishing requires

$$0 \le m + jN \le L,$$

while the second requires

$$0 \le m - 1 + jN \le L.$$

The smallest possible index in the union of these two supports is

$$-\left\lfloor \frac{m}{N} \right\rfloor,$$

coming from the first coefficient. The largest is

$$1 + \left\lfloor \frac{m-1}{N} \right\rfloor,$$

coming from the second. If N divides m, the lower endpoint $j = -m/N$ contributes $\binom{L}{0} = 1$, so it cannot be discarded. □

The bounds printed in Remark 4.3 are

$$-\left\lfloor \frac{m-1}{N} \right\rfloor \le j \le 1 + \left\lfloor \frac{m}{N} \right\rfloor.$$

When $N \nmid m$, these describe the same effective support as equation (6). When $N \mid m$, however, the printed lower bound omits the nonzero term $j = -m/N$, while the printed upper bound merely adds a zero term. The net omission is exactly 1.

Proposition 6.2 (Short range and equality boundary)**.** *If $1 \le m < N$, only $j = 0$ and $j = 1$ contribute to equation (5), and the result is equation (7). If $m = N$, the additional index $j = -1$ contributes 1, and the result is equation (8).*

Proof. For $m < N$, equation (6) reduces to $0 \le j \le 1$. Therefore

$$\begin{aligned}\|P_n\|_{X,2m}^{2m} &= \binom{L}{m} - \binom{L}{m-1} + \binom{L}{m+N} - \binom{L}{m-1+N} \\ &= \binom{L}{m} - 2\binom{L}{m-1} + \binom{L}{m-2},\end{aligned}$$

where the second line uses binomial symmetry and $L = N - 2 + 2m$. Factoring out $\binom{L}{m}$ gives

$$\binom{L}{m}\frac{n^2+n-2m}{(n+m)(n+m+1)}.$$

The identity

$$\frac{\binom{L}{m}}{(n+m)(n+m+1)} = \frac{\binom{n+2m+1}{m}}{(n+2m)(n+2m+1)}$$

then yields equation (7).

At $m = N$, equation (6) gives $-1 \le j \le 1$. The $j = -1$ term is

$$\binom{L}{0} - \binom{L}{-1} = 1,$$

while the $j = 0, 1$ terms simplify exactly as above. Substituting $m = n + 1$ produces equation (8). □

Table 1: Audit of the equality case $m = n + 1$. The non-strict short formula is lower than the exact value by one in every displayed case.

n	$d = 2(n+1)$	exact value	non-strict short RHS	corrected RHS
1	4	0	−1	0
2	6	1	0	1
3	8	16	15	16
4	10	144	143	144
5	12	1093	1092	1093

7 Rational generating function and recurrence

The endpoint-walk identity also yields a compact evaluation mechanism across all even degrees. Define

$$a_{n,m} = \|P_n\|_{X,2m}^{2m} \quad (m \ge 1), \qquad a_{n,0} = 1.$$

Theorem 7.1 (Generating function)**.** *For every $n \ge 1$,*

$$\sum_{m\ge 0} a_{n,m} z^m = \left(\sum_{r=0}^{\lfloor n/2\rfloor} (-1)^r \binom{n-r}{r} z^r\right)^{-1}. \tag{14}$$

Equivalently, for $m \ge 1$,

$$a_{n,m} = \sum_{r=1}^{\min(m,\lfloor n/2\rfloor)} (-1)^{r+1}\binom{n-r}{r} a_{n,m-r}. \tag{15}$$

Proof. The path determinant $D_n(t) = \det(I - tA_n)$ obeys the continuant recurrence

$$D_n(t) = D_{n-1}(t) - t^2 D_{n-2}(t), \qquad D_0(t) = D_1(t) = 1.$$

Induction gives

$$D_n(t) = \sum_{r=0}^{\lfloor n/2 \rfloor} (-1)^r \binom{n-r}{r} t^{2r}. \tag{16}$$

By equation (2), $D_n(t)^{-1}$ is the CHS generating function. All odd coefficients vanish, so substituting $z = t^2$ into equation (16) gives equation (14). Multiplying both sides by the denominator and comparing coefficients yields equation (15). □

Table 2: Low-order generating functions and the first values $a_{n,0}, \dots, a_{n,6}$.

n	$\sum_{m\geq 0} a_{n,m} z^m$	first values
1	1	1, 0, 0, 0, 0, 0, 0
2	$(1-z)^{-1}$	1, 1, 1, 1, 1, 1, 1
3	$(1-2z)^{-1}$	1, 2, 4, 8, 16, 32, 64
4	$(1-3z+z^2)^{-1}$	1, 3, 8, 21, 55, 144, 377
5	$(1-4z+3z^2)^{-1}$	1, 4, 13, 40, 121, 364, 1093
6	$(1-5z+6z^2-z^3)^{-1}$	1, 5, 19, 66, 221, 728, 2380

For example, $a_{4,m} = F_{2m+2}$ in terms of the Fibonacci sequence, and $a_{5,m} = (3^{m+1} - 1)/2$. More generally, equation (14) shows that for each fixed n, the even-degree CHS powers form an integer linear-recurrence sequence.

8 Exact verification

The proof is symbolic and does not depend on numerical evidence. The accompanying verification script nevertheless audits the formulas by independent finite procedures. For every pair

$$1 \leq n \leq 24, \qquad 0 \leq m \leq 48,$$

it compares:

1. dynamic programming for endpoint walks in P_n;
2. coefficient inversion of the determinant polynomial in equation (16);
3. the finite reflection sum in equation (5) using the corrected support;
4. the short formula throughout the strict range $1 \leq m < n + 1$;
5. the corrected equality formula at $m = n + 1$.

All 1,176 parameter pairs agree exactly. Only integer arithmetic is used for the walk, determinant, and reflection calculations; rational arithmetic is used for the displayed closed forms.

Independent review targets

A specialist review can be concentrated on four points: the endpoint cofactor in lemma 3.1; the coefficient shift in theorem 3.2; the image-kernel boundary cancellations in lemma 5.1; and the support endpoints in proposition 6.1. Once those are confirmed, the conjectured formulas and the boundary correction follow mechanically.

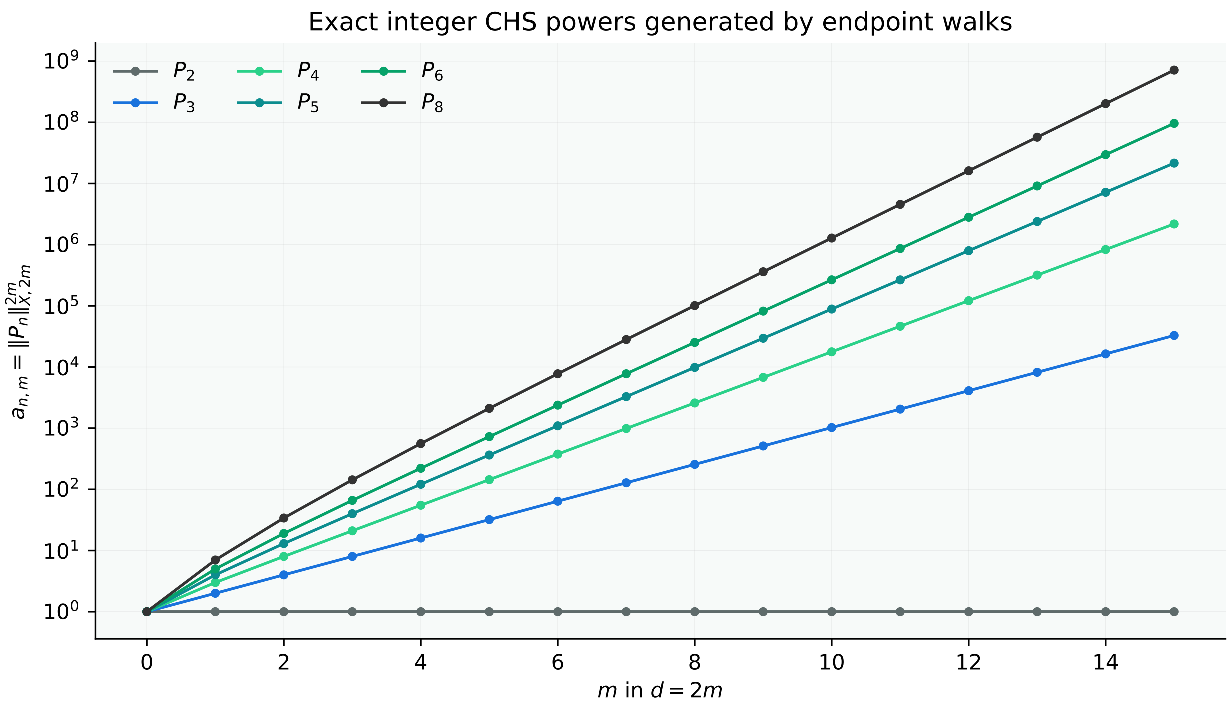

Figure 3: Exact values $a_{n,m} = \|P_n\|_{X,2m}^{2m}$ for selected path orders. The integrality is explained by the endpoint-walk interpretation, while the exponential growth rates are governed by the path eigenvalues.

9 Discussion

The resolution is short because the CHS generating function and the path resolvent share the same determinant. For a general graph, a resolvent entry carries a cofactor depending on the selected vertices. The path is special: the endpoint cofactor is the monomial t^{n-1}. That monomial shift makes every CHS coefficient equal to one endpoint-walk count.

This viewpoint produces three layers of evaluation:

- the resolvent identity gives the structural interpretation;
- path eigenvectors give the finite trigonometric formula;
- the reflection principle gives the finite binomial formula and exposes the support boundary.

It also explains several features that are not obvious from the original conjecture: integrality, nonnegativity, the vanishing of odd degrees, the recurrence in m, and the exact source of the equality-case discrepancy.

Priority and review caveat

The mathematical derivation is exact, but novelty has not been independently certified. The endpoint-resolvent identity may be implicit in literature on continuants, Jacobi matrices, lattice paths, or symmetric functions even if it has not previously been applied to this stated open problem. Until specialists confirm both the proof and prior art, this document should be cited as a candidate resolution with a proposed correction to the auxiliary boundary statements in Remark 4.3.

Reproducibility statement

The source package accompanying this preprint contains the complete LuaLaTeX manuscript, formula-derived cover and figures, the exact verification script, recorded verification output, and release checksums. The verification script requires only the Python standard library.

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Generative artificial intelligence and other computational tools were used as research instruments to support construction search, symbolic and numerical analysis, literature review, proof development, verification, and manuscript preparation. Their use does not constitute independent validation of the results presented.

This manuscript has not yet undergone independent peer review and should not be regarded as an established resolution of the stated problem unless and until its arguments have been examined and confirmed by qualified mathematicians. Independent verification, criticism, replication, and identification of errors are expressly invited.

Data, code, funding, and competing interests

All source and verification materials are included with the release. No external dataset was used. This work received no external funding specifically designated for the result. The issuing organization declares no competing financial interest in the mathematical conclusion.

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